CENGAGE | WEBASSIGN

futchhd.06@tamu.edu
(Sign out)

Home My Assignments Grades
Communication Calendar
My eBooks

Math 151, section 510-518, Fall 2024

Math 151 Homework Section 3.1 (Homework)

INSTRUCTOR
Sang Lee
Texas A&M University

Current Score

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
POINTS	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5	–/1.5

TOTAL SCORE

–/30 0.0%

Due Date

WED, SEP 25, 2024
11:55 PM CDT

Request Extension

Assignment Submission & Scoring

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your best submission for each question part is used for your score.

1. **[–/1.5 Points]** DETAILS MY NOTES

Differentiate these functions.

(a) $f(x) = 3\pi^4$

$f'(x) =$

(b) $g(x) = 2^{90}$

$g'(x) =$

(c) $h(x) = e^{75}$

$h'(x) =$

2. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.XP.1.016.**

Differentiate the function.

$$f(x) = x^3 - 4x + 3$$

$f'(x) =$

3. **[–/1.5 Points]** DETAILS MY NOTES

Differentiate the function.

$$h(x) = \frac{8}{9x} - \frac{7}{x^6} + \frac{8}{x^9}$$

$h'(x) =$

4. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.XP.1.027.**

Differentiate the following function.

$$u = \sqrt[7]{t^2} + 2\sqrt{t^3}$$

$u'(t) =$

5. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.021.MI.**

Differentiate the function.

$$y = 2e^x + \frac{7}{\sqrt[3]{x}}$$

$y' =$

6. **[–/1.5 Points]** DETAILS MY NOTES

Differentiate the function.

$$f(x) = \frac{x^8 + 4x^4 + 10x}{x^4}.$$

$f'(x) =$

7. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.019.MI.**

Differentiate the function.

$$f(x) = x^3(x + 4)$$

$f'(x) =$

8. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.050.**

Find the first and second derivative of the function.

$$G(r) = \sqrt{r} + \sqrt[7]{r}$$

$G'(r) =$

$G''(r) =$

9. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.035.**

Consider the following.

$$y = tx^2 + t^5x$$

Find $\frac{dy}{dx}$.

$\frac{dy}{dx} =$

Find $\frac{dy}{dt}$.

$\frac{dy}{dt} =$

10. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.053.**

The equation of motion of a particle is $s = t^3 - 27t$, where s is measured in meters and t is in seconds. (Assume $t \geq 0$.)

(a) Find the velocity and acceleration as functions of t.

$v(t) =$

$a(t) =$

(b) Find the acceleration, in m/s^2, after 4 seconds.

______ m/s^2

(c) Find the acceleration, in m/s^2, when the velocity is 0.

______ m/s^2

11. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.XP.1.031.**

Find an equation of the tangent line to the curve at the given point.

$$y = x^4 + 7x^2 - x, \ (1, 7)$$

$y =$

12. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.038.**

Find an equation of the tangent line to the curve at the given point.

$$y = 5e^x + x, \ (0, 5)$$

$y =$

13. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.059.**

Find the points on the curve $y = x^3 + 3x^2 - 9x + 3$ where the tangent is horizontal.

smaller x-value $(x, y) = \Big(\quad \Big)$

larger x-value $(x, y) = \Big(\quad \Big)$

14. **[–/1.5 Points]** DETAILS MY NOTES

Determine the value(s) of x where the function will have an instantaneous rate of change of 9. (Enter your answers as a comma-separated list. If an answer does not exist, enter DNE.)

$$f(x) = x^3 + 21x^2 + 9x + 46$$

$x =$

15. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.XP.1.038.**

Find an equation of the tangent line to the curve $y = x\sqrt{x}$ that is parallel to the line $y = 6 + 6x$.

$y =$

16. **[–/1.5 Points]** DETAILS MY NOTES

Find a parabola with equation $y = ax^2 + bx + c$ that has slope -7 at $x = 0$, slope 17 at $x = 2$, and passes through the point (1, 2).

$y =$

17. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.081.**

For what values of a and b is the line $2x + y = b$ tangent to the parabola $y = ax^2$ when $x = 4$?

$a =$

$b =$

18. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.068.**

Find equations of both lines through the point $(2, -3)$ that are tangent to the parabola $y = x^2 + x$.

smaller slope $y =$

larger slope $y =$

19. **[–/1.5 Points]** DETAILS MY NOTES

(a) Consider the function $f(x) = |x^2 - 16|$

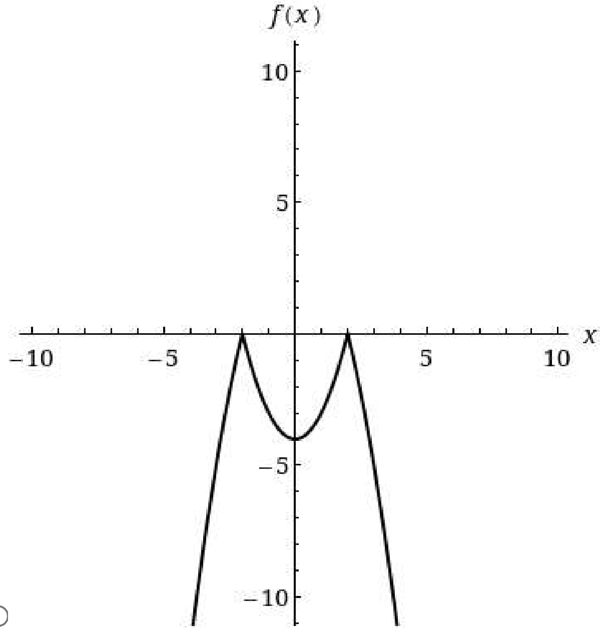
Find a formula for f'.

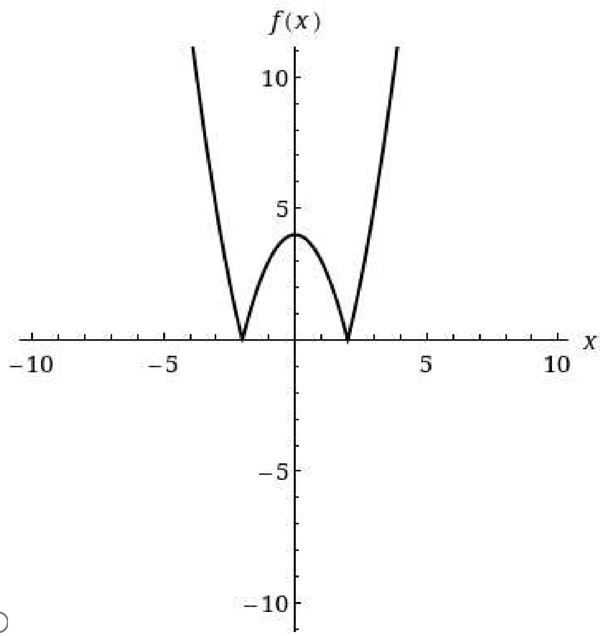
$f'(x) = \begin{cases} \_\_\_\_ & \text{if } |x| > \_\_\_\_ \\ \_\_\_\_ & \text{if } |x| < \_\_\_\_ \end{cases}$

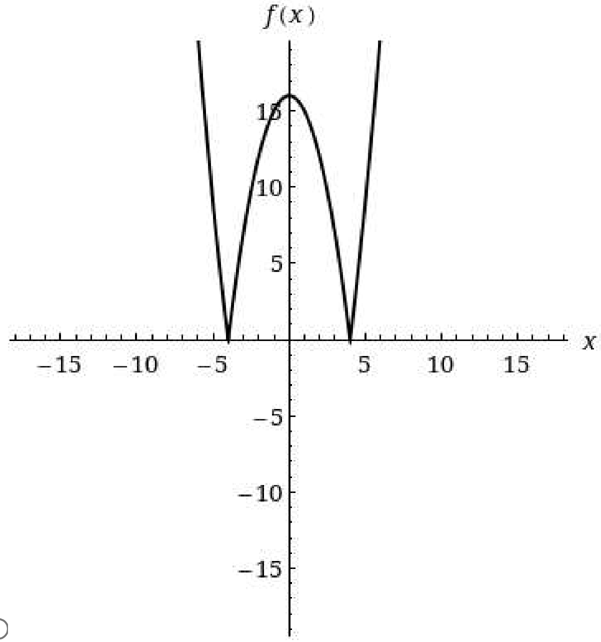
(b) For what values of x is the function not differentiable? (Enter your answers as a comma-separated list.)

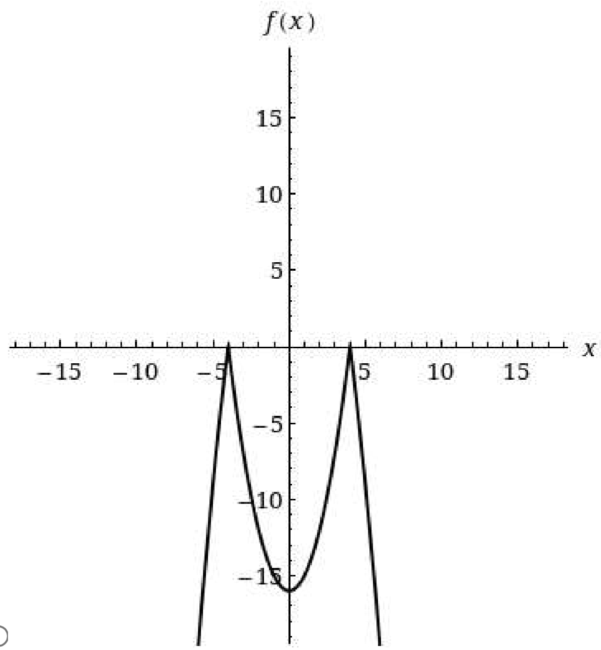
$x =$

(c) Sketch the graph of f.

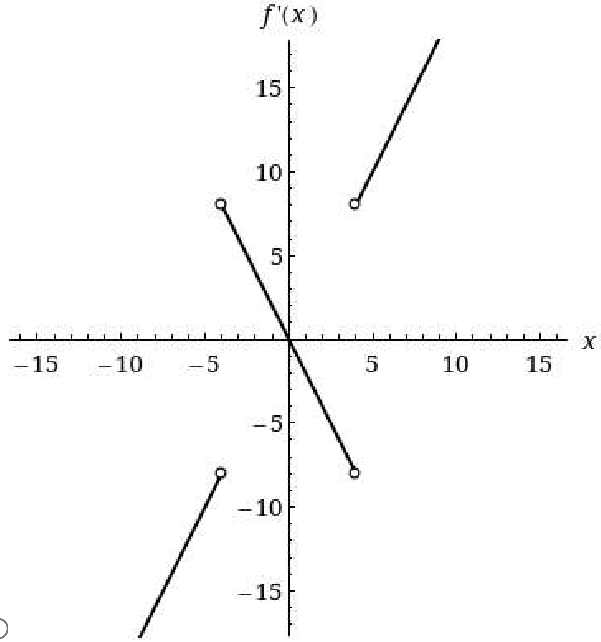

f(x)
10
5
x
−10
−5
5
10
−5
−10


f(x)
10
5
x
−10
−5
5
10
−5
−10

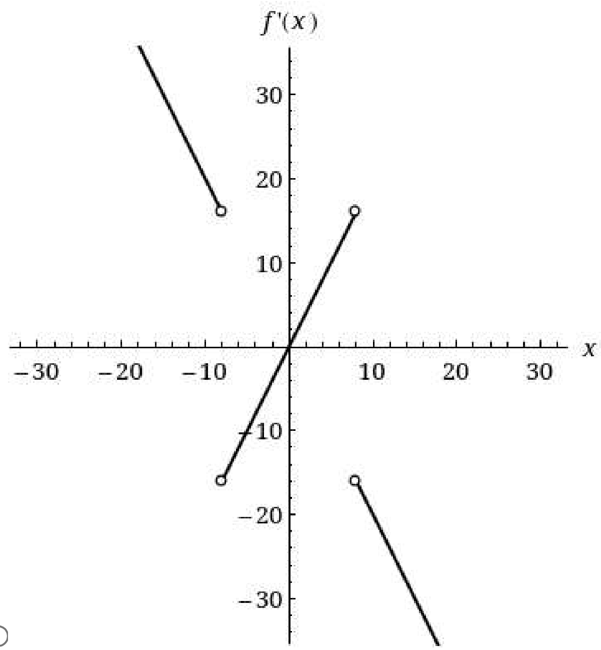




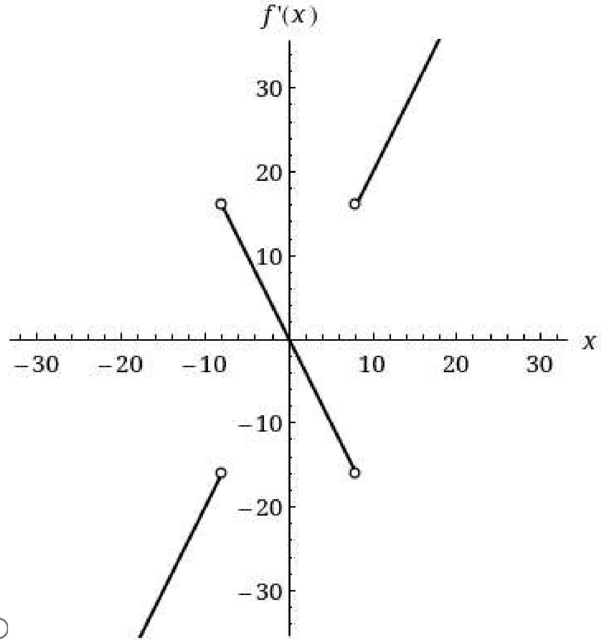
Sketch the graph of f'.

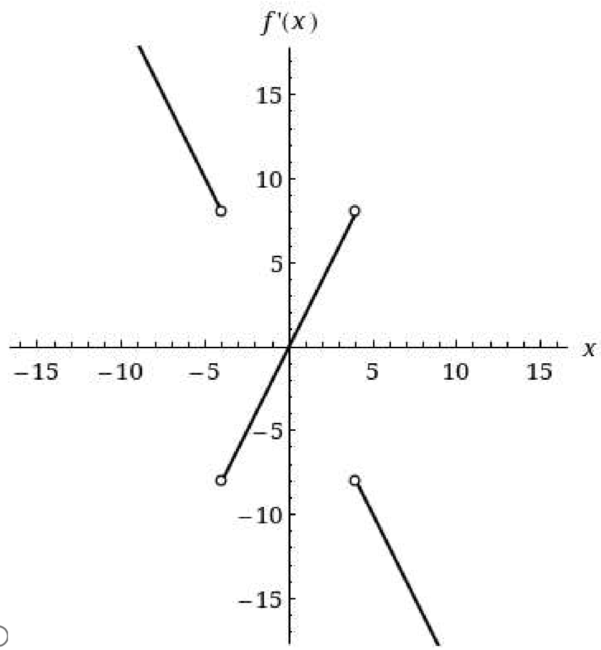


f'(x)
15
10
5
−5
−10
−15
−15
−10
−5
5
10
15
x



f'(x)
30
20
10
−10
−20
−30
−30
−20
−10
10
20
30
x





20. **[–/1.5 Points]** DETAILS MY NOTES **SCALCET9 3.1.085.**

Find the values of m and b that make f differentiable everywhere.

$$f(x) = \begin{cases} x^2 & \text{if } x \le 8 \\ mx + b & \text{if } x > 8 \end{cases}.$$

m =

b =

Home My Assignments Request Extension